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Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex:6 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim:

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacing.

Objective:

- 1.To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.
- 2.To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.
- 3.To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix. Software Required
- 4.MATLAB

Procedure

- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

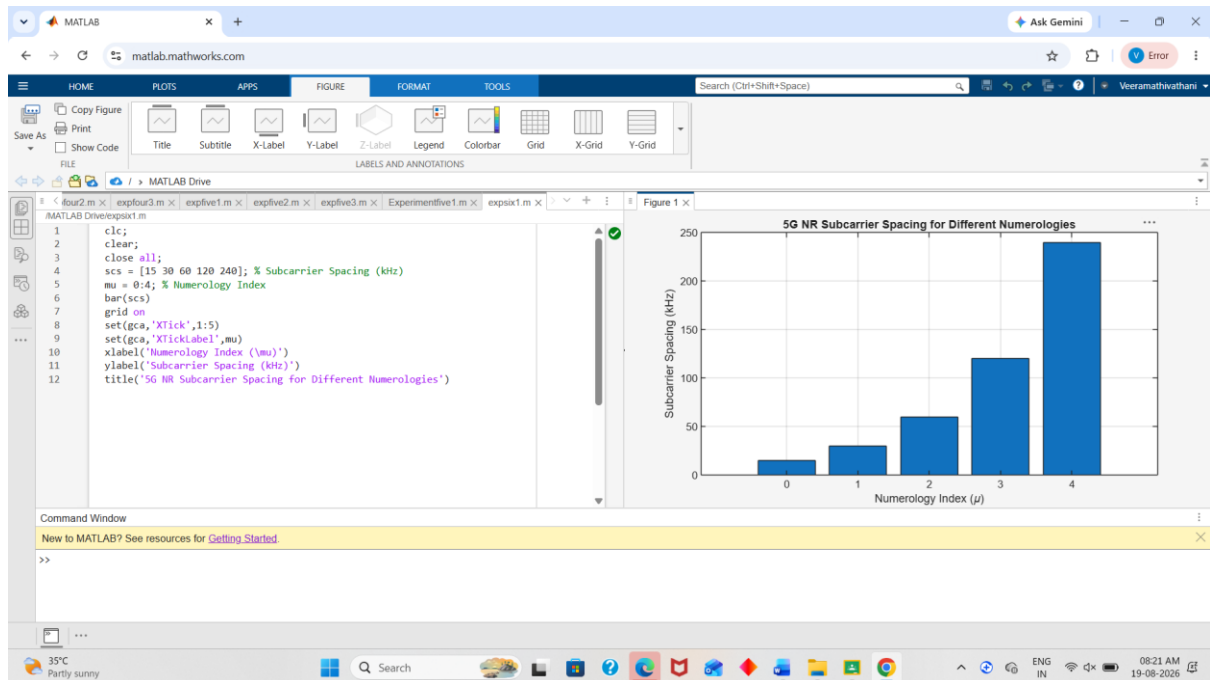
Objective 1: Matlab Code and Output

```
clc;  
clear;  
close all;  
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)  
mu = 0:4; % Numerology Index  
bar(scs)  
grid on
```

```

set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')

```



Objective 2: Matlab Code and Output

```

clear;

close all;

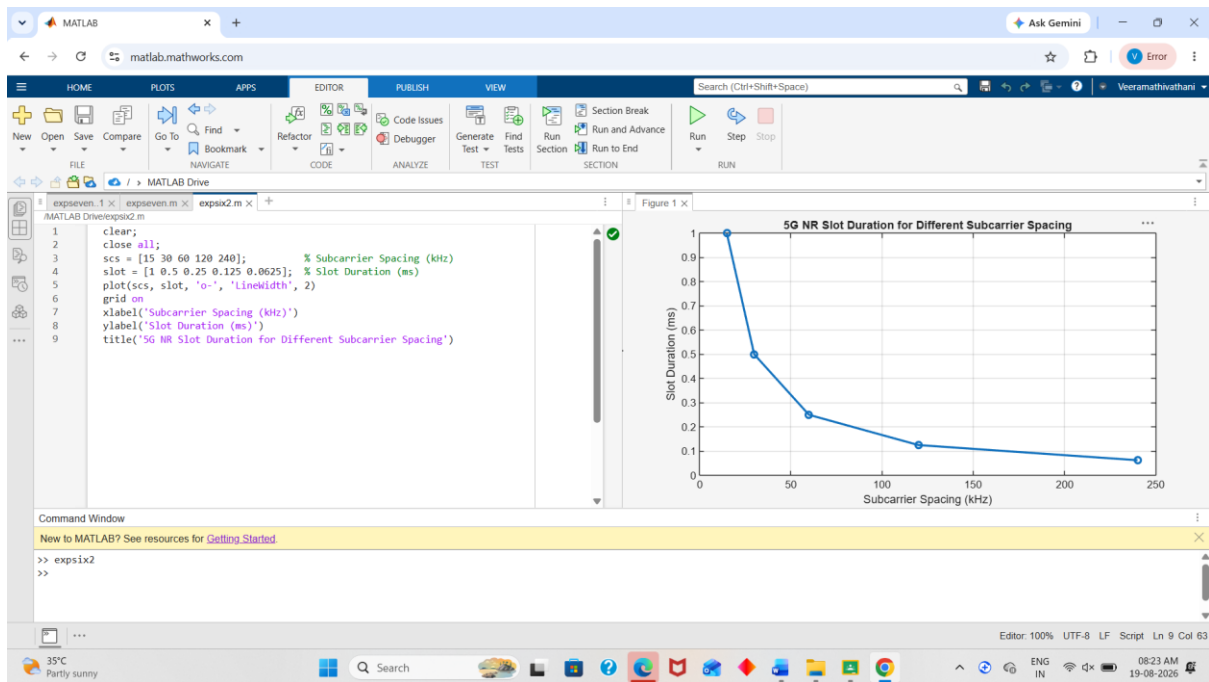
scs = [15 30 60 120 240];      % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)

grid on

xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')

```



Objective 3: Matlab Code and Output

clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot

figure

bar(scs, symbols)

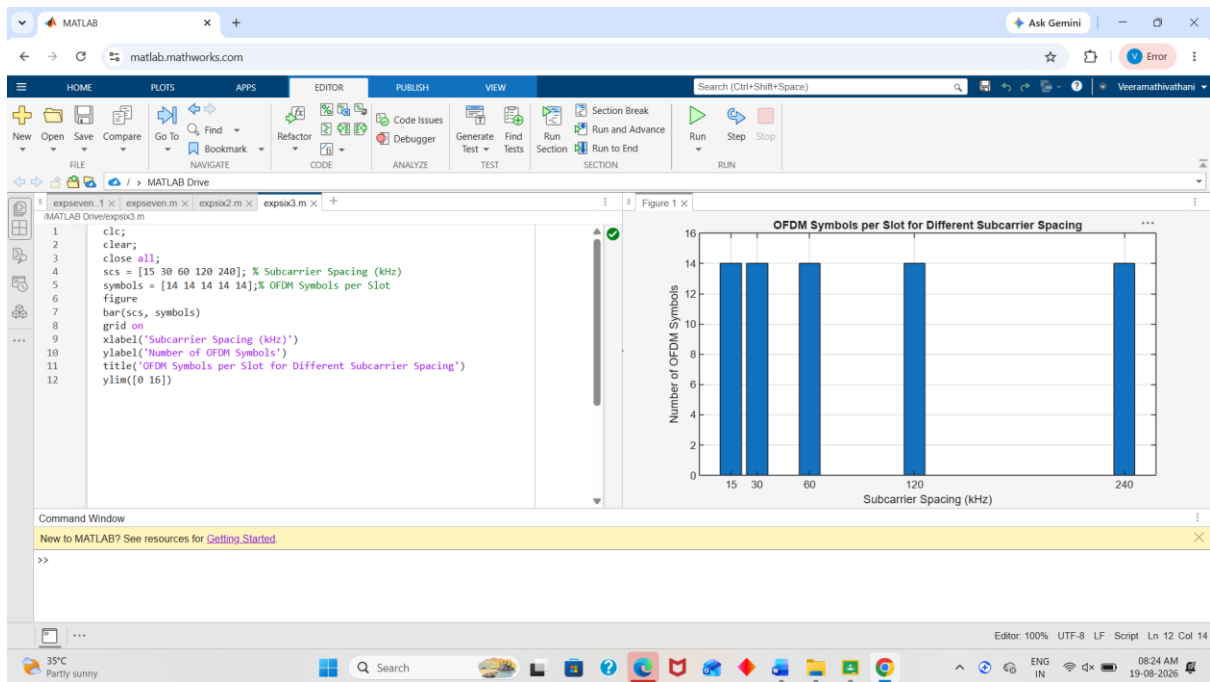
grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.